16届省赛（第一套）润色版代码

```c
#include <STC15F2K60S2.H>
#include "led.h"
#include "init.h"
#include "key.h"
#include "seg.h"
#include "onewire.h"
#include "iic.h"
#include "ultrasound.h"
#include "math.h"
idata unsigned char Key_Val,Key_Old,Key_Down,Key_Up;
idata unsigned char Key_Slow_Down;
idata bit Long_Press_Flag;//按键长按标志位
idata unsigned int Long_Press_Time_2S;//按键长按2s计时器

idata unsigned char Seg_Pos;
pdata unsigned char Seg_Buf[]={10,10,10,10,10,10,10,10};
idata unsigned char Seg_Slow_Down;
idata unsigned char Seg_Disp_Mode=0;//0-环境状态 1-运动检测 2-参数设置 3-统计数据
idata unsigned char Para_Disp_Mode;//参数设置界面的两个界面 0-温度参数 1-距离参数
idata unsigned char Temperature_Val_Para=30;//温度参数
idata unsigned char Distance_Para=30;//距离参数

idata bit Relay_Led_Lock;//继电器，led状态锁定标志位
idata bit Relay_Old;//继电器从关到开的标志位
idata unsigned int Relay_Count;//继电器吸合次数
pdata unsigned char ucled[]={0,0,0,0,0,0,0,0};
idata bit Led_Flash_Flag;//1-亮 0-灭
idata unsigned char Led_Flash_Clock;//led闪烁计时器 100ms
```

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idata unsigned char Temperature_Val;//环境温度
idata unsigned int Temperature_Slow_Down;
idata bit Htemperature_Flag;//高温标志位 0-未得到高温 1-高温

idata unsigned char Light_Level;//光强等级
idata unsigned char AD_Slow_Down;

idata unsigned char Motion_State; //运动状态 0-静止 1-徘徊 2-跑动
idata bit Run_State;//跑动的状态
idata bit Stay_State;//静止状态
idata bit Paralysis_State;//徘徊状态
idata unsigned char Distance_Val;
idata unsigned char Distance_Old;
idata unsigned char Distance_Sub;//距离差值
idata unsigned int Distance_Slow_Down;
idata bit First_Get_Distance;//当从0->1表示已经获取里第一次的距离
idata bit State_Change_Flag;//运动状态锁定标志位 1-锁定 0-未锁定
idata unsigned int Distance_Lock_3S;//运动状态锁定3s计时器
idata bit Distance_Flag;//距离接近标志位 1-接近



void Key_Proc()
{
    if(Key_Slow_Down<10) return;
    Key_Slow_Down=0;
    Key_Val=Key_Read();
    Key_Down=Key_Val&(Key_Val^Key_Old);
    Key_Up=~Key_Val&(Key_Val^Key_Old);
    Key_Old=Key_Val;
    if(Key_Down==4)
    {
        Seg_Disp_Mode=(++Seg_Disp_Mode)%4;//0-3之间循环
        Para_Disp_Mode=0;//让参数界面默认置为温度参数子界面
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        if(Seg_Disp_Mode==2)//进入参数界面，锁定
        {
            Relay_Led_Lock=1;
        }
        else//退出参数界面，取消锁定
        {
            Relay_Led_Lock=0;
        }
    }
    //在参数界面下
    if(Seg_Disp_Mode==2)
    {
        if(Key_Down==5)
        {
            Para_Disp_Mode=(++Para_Disp_Mode)%2;
        }
        if(Key_Down==8)
        {
            if(Para_Disp_Mode==0)
            {
                if(Temperature_Val_Para<80)
                    Temperature_Val_Para++;
            }
            else
            {
                if(Distance_Para<80)
                    Distance_Para+=5;
            }
        }
        if(Key_Down==9)
        {
            if(Para_Disp_Mode==0)
            {
                if(Temperature_Val_Para>20)
                    Temperature_Val_Para--;
            }
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                else
                {
                    if(Distance_Para>20)
                        Distance_Para-=5;
                }
            }
        }
        //在统计界面下
        if(Seg_Disp_Mode==3)
        {
            if(Long_Press_Time_2S>=2000)
                Relay_Count=0;//清零继电器吸合次数
            if(Key_Old==89)//按键89同时按下
            {
                Long_Press_Flag=1;
            }
            else//松手或未同时按下
            {
                Long_Press_Flag=0;//标志位置0
                Long_Press_Time_2S=0;//计时器归0
            }
        }
    }

    void Seg_Proc()
    {
        if(Seg_Slow_Down<80) return;
        Seg_Slow_Down=0;
        switch(Seg_Disp_Mode)
        {
            case 0:
                Seg_Buf[0]=11;
                Seg_Buf[1]=Temperature_Val/10%10;
                Seg_Buf[2]=Temperature_Val%10;
                Seg_Buf[3]=Seg_Buf[4]=Seg_Buf[5]=10;
                Seg_Buf[6]=12;
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            Seg_Buf[7]=Light_Level;
            break;
        case 1:
            Seg_Buf[0]=13;
            Seg_Buf[1]=Motion_State+1;
            Seg_Buf[2]=Seg_Buf[3]=Seg_Buf[4]=10;
            Seg_Buf[5]=Distance_Val/100%10;//直接按正常的写就是
高位补0，如56/100=0 0%10=0
            Seg_Buf[6]=Distance_Val/10%10;
            Seg_Buf[7]=Distance_Val%10;
            break;
        case 2:
            Seg_Buf[0]=14;
            Seg_Buf[2]=Seg_Buf[3]=Seg_Buf[4]=Seg_Buf[5]=10;
            if(Para_Disp_Mode==0)
            {
                Seg_Buf[1]=11;
                Seg_Buf[6]=Temperature_Val_Para/10%10;
                Seg_Buf[7]=Temperature_Val_Para%10;
            }
            else
            {
                Seg_Buf[1]=13;
                Seg_Buf[6]=Distance_Para/10%10;
                Seg_Buf[7]=Distance_Para%10;
            }
            break;
        case 3:
            Seg_Buf[0]=12;
            Seg_Buf[1]=11;
            Seg_Buf[2]=Seg_Buf[3]=10;
            Seg_Buf[4]=(Relay_Count>=1000)?
Relay_Count/1000%10:10;
            Seg_Buf[5]=(Relay_Count>=100)?
Relay_Count/100%10:10;
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            Seg_Buf[6]=(Relay_Count>=10)?
Relay_Count/10%10:10;
            Seg_Buf[7]=Relay_Count%10;
            break;
    }
}

void Led_Proc()
{
    if(!Relay_Led_Lock)//如果不锁定
    {
        if(Distance_Flag)
        {
            switch(Light_Level)
            {
                case 1:
                    ucled[0]=1;
                    ucled[1]=ucled[2]=ucled[3]=0;
                    break;
                case 2:
                    ucled[0]=ucled[1]=1;
                    ucled[2]=ucled[3]=0;
                    break;
                case 3:
                    ucled[0]=ucled[1]=ucled[2]=1;
                    ucled[3]=0;
                    break;
                case 4:
                    ucled[0]=ucled[1]=ucled[2]=ucled[3]=1;
                    break;
            }
        }
        else
        {
            ucled[0]=ucled[1]=ucled[2]=ucled[3]=0;
        }
```

```
            if(Distance_Flag==1&&Htemperature_Flag==1)
            {
                Relay(1);
                if(Relay_Old==0)
                {
                    Relay_Old=1;
                    Relay_Count++;
                }
            }
            else
            {
                Relay(0);
                Relay_Old=0;
            }
        }
        if(Stay_State)
            ucled[7]=0;
        else if(Paralysis_State)
            ucled[7]=1;
        else if(Run_State)
            ucled[7]=Led_Flash_Flag;
    }
    void Get_AD()
    {
        float temp;
        if(AD_Slow_Down<100)return;
        AD_Slow_Down=0;
        temp=AD_Read(0x01)/51.0;
        if(temp>=3.0)Light_Level=1;
        else if(temp>=2.0)Light_Level=2;
        else if(temp>=0.5)Light_Level=3;
        else Light_Level=4;
    }

    void Get_Temperature()
    {
```

```
    if(Temperature_Slow_Down<300)return;
        Temperature_Slow_Down=0;
    Temperature_Val=(unsigned char)rd_temperature();;
    if(Temperature_Val>Temperature_Val_Para)
    {
        Htemperature_Flag=1;
    }
    else
    {
        Htemperature_Flag=0;
    }

}

void Get_Distance()
{
    unsigned char temp_motion_state;
    if(Distance_Slow_Down<999) return;
        Distance_Slow_Down=0;
    if(First_Get_Distance==1)
    {
        Distance_Val=Ut_Wave_Data();
        Distance_Sub=abs(Distance_Val-Distance_Old);
        Distance_Old=Distance_Val;
        Distance_Flag=(Distance_Val<Distance_Para);
        if(!State_Change_Flag)
        {
            if(Distance_Sub<5) temp_motion_state=0;
            else if(Distance_Sub<10) temp_motion_state=1;
            else temp_motion_state=2;

            //如果状态改变
            if(temp_motion_state!=Motion_State)
            {
                State_Change_Flag=1;
                Distance_Lock_3S=0;   //3s锁定计时器初始化为0
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                Motion_State=temp_motion_state;
            }

        }

        if(!Relay_Led_Lock)//led和继电器不锁定才根据实际情况把下面的三个状态置0或置1
        {
            Stay_State=(Motion_State==0);
            Paralysis_State=(Motion_State==1);
            Run_State=(Motion_State==2);
        }
    }
    else
    {
        First_Get_Distance=1;//获取了第一次的距离
        Distance_Val=Ut_Wave_Data();
        Distance_Old=Distance_Val;
    }

}
void Timer1_Init(void)      //1毫秒@12.000MHz
{
    AUXR &= 0xBF;           //定时器时钟12T模式
    TMOD &= 0x0F;           //设置定时器模式
    TL1 = 0x18;             //设置定时初始值
    TH1 = 0xFC;             //设置定时初始值
    TF1 = 0;                //清除TF1标志
    TR1 = 1;                //定时器1开始计时
    ET1 = 1;                //使能定时器1中断
    EA=1;             //打开全局中断
}

void Timer1_Isr(void) interrupt 3
{
    Key_Slow_Down++;
```

```
    Seg_Slow_Down++;
    Temperature_Slow_Down++;
    Distance_Slow_Down++;
    AD_Slow_Down++;

    //数码管动态扫描
    if(Seg_Buf[Seg_Pos]>20)
        Seg_Disp(Seg_Pos,Seg_Buf[Seg_Pos]-',',1);
    else
        Seg_Disp(Seg_Pos,Seg_Buf[Seg_Pos],0);
    Seg_Pos=(++Seg_Pos)%8;


    if(Long_Press_Flag)//如果长按按键89
    {
        if(++Long_Press_Time_2S>=2000)
Long_Press_Time_2S=2001;
    }
    else//如果没有长按
    {
        Long_Press_Time_2S=0;
    }

    if(Run_State)
    {
        if(++Led_Flash_Clock==100)
        {
            Led_Flash_Clock=0;
            Led_Flash_Flag=!Led_Flash_Flag;//在0-1之间翻转
        }
    }
    else
    {
        Led_Flash_Clock=0;
        Led_Flash_Flag=0;
    }
```

```
    if(State_Change_Flag)
    {
        if(++Distance_Lock_3S==3000)
        {
            Distance_Lock_3S=0;
            State_Change_Flag=0;
        }
    }
    Led_Disp(ucled);
}


void main()
{
    while(rd_temperature()==85.0);
    AD_Read(0x01);
    System_Init();
    Timer1_Init();
    while(1)
    {
        Key_Proc();
        Seg_Proc();
        Led_Proc();
        Get_AD();
        Get_Temperature();
        Get_Distance();
    }
}
```